

Contact

Rio de Janeiro
+5522992784262 (Mobile)
gustavoj.m.aguiar@outlook.com

www.linkedin.com/in/gjmaguiar
(LinkedIn)
www.researchgate.net/profile/Gustavo_Aguiar4 (Personal)
lattes.cnpq.br/1848857688296947
(Other)
orcid.org/0000-0002-0538-0845
(Other)

Top Skills

Engenharia de dados
SAS Viya
Experiência do usuário (UX)

Languages

Português (Native or Bilingual)
Inglês (Full Professional)
Espanhol (Professional Working)

Certifications

Formação Master em MS Excel - 2.0
Machine Learning Using SAS Viya
SAS Visual Analytics 1 for SAS Viya: Basics
SAS Programming 3: Advanced Techniques
Certificado Planejamento Estratégico

Honors-Awards

Medalha de Bronze na Olimpíada Brasileira de Informática
Prêmio Láurea Acadêmica - 2020

Publications

Application of FMEA tool in a driver-training centre
Construção de dicionário de variáveis de monitoramento do sistema BOP para auxiliar implantação de CBM

Gustavo Aguiar, M.Sc.

Senior Data Scientist | Machine Learning Engineer | Solutions Architect | SAS Viya Specialist | AI | Cloudera Specialist | NLP | MLOps | Computer Vision
Rio de Janeiro, Brazil

Summary

Senior Data Scientist & ML Engineer – Turning data into products that create real impact, especially by accelerating business processes. I plan and design solutions, build data-science and ML-engineering products, and mentor professionals in the field.

What I've been doing:

1. State Governments – Espírito Santo (SEGER, SESA, SEDU, SEJUS, SESP, SECONT, Civil Police, DETRAN-ES) & Rio de Janeiro (SEPOL-RJ, DETRAN-RJ)

- Designed and implemented Data Lakehouses with Cloudera Data Platform (HDFS & Impala) + PySpark + Airflow. Thousands of sources (Contracts, Grants, Health, Education, Police Incidents, and more) are now centralized, monitored, and alert-enabled.

- Delivered BI layers in SAS Viya and AI models via MLflow/Viya: fare-price anomaly detection, judicial-document classifiers, contract-entity extraction, procurement alerts, warrant triage, and forecasting for energy and water use.

- Built data apps on Kubernetes (Rancher) with Python + Node/React, integrating CDP and SAS Viya through REST APIs.

2. Eletrobras Group

- Built a legal Data Mart in SAS Viya for Furnas, Chesf, CGT Eletrosul, and Eletronorte.

- Orchestrated ingestion and refresh pipelines feeding 15+ management and operational dashboards.

Propositions of innovation practices in a software company based on the maturity assessment of innovation management and literature good practices

Grey Theory applied to sales forecasting in a building materials company

Teoria Grey Aplicada na Previsão da Demanda em uma Loja de Materiais de Construção In: Lean Six Sigma: Iniciativas para a Melhoria Contínua em Operações e Serviços

- Modeled forecasts of case volumes, likelihood of unfavorable sentences, and early case-closure detection, shrinking legal liabilities.

3. Oil & Gas (Petrobras, Constellation, Transocean)

- Developed a BOP simulator, BI environment for 100+ sensor tags, and an anomaly-alert system based on CBM methodology.

Techs: Cloudera DP | HDFS | Impala | PySpark | Airflow | SAS Viya | LLM | MLflow | Docker/Kubernetes (Rancher) | AWS | Python | Power BI | Node.js | React | MLOps | Digital Twin | Computer Vision | NLP | spaCy | NLTK

gustavoj.m.aguiar@outlook.com | +55 22 99278-4262

Experience

Vert TI

3 years 6 months

Senior Data Scientist

July 2022 - Present (3 years 1 month)

Brasília, Federal District, Brazil

- Developed automated data pipelines and ETL processes, empowering dashboards with updated information and facilitating the construction and execution of machine learning models.
- Designed various machine learning models, enhancing analytical dashboards with features such as detection of legal process closure via NLP, projection of process volume, and process classification based on textual content, among other models.
- Managed the deployment and operationalization of machine learning models, ensuring their full functionality.
- Designed and constructed legal intelligence dashboards, facilitating strategic data-based decision-making.
- Served as a mentor for team colleagues, guiding them in developing machine learning models and data pipelines.
- As Technical Reference, I manage the planning and execution of the team's main activities.

- Promoted regular meetings for tracking, clarifying doubts, and fostering collaboration within the team.
- Developed the data architecture in SAS Viya, Cloudera and Kubernetes (K8S), optimizing organization and access to data.
- Built automated data pipelines and ETL processes using Airflow and PySpark or SAS Compute for constantly updating data sources in all layers of a Data Lake (Bronze, Silver, Gold) [HDFS Cloudera] and Data Warehouse [Impala].
- Guided the adoption of code versioning practices and continuous integration / continuous delivery (CI/CD).
- Responsible for defining the architecture of the analytical solution and guiding the main technical decisions of the project.
- Designed dashboards for payroll auditing, enhancing the transparency and efficiency of the process.

Data Scientist

February 2022 - June 2022 (5 months)

Brasília, Federal District, Brazil

I was part of the development team for an Analytics and Business Intelligence (BI) project to optimize the Brazilian Court of Auditors audit process. My responsibilities and achievements include the following:

- Implementation of data pipelines and ETL (Extract, Transform, Load) processes for efficient data collection, processing, and analysis.
- Development of Machine Learning models in Natural Language Processing (NLP) for Named Entity Recognition (NER), leading to the automatic identification of entities in documents (for example, public bidding announcements).
- Design and implementation of automatic processes for fraud detection by verifying various data sources, aiding in preventing fraudulent activities.
- Construction of dashboards to present a clear view of the data, which is crucial for guiding strategic decisions and assisting the client in the decision-making process during the audit.

Universidade Federal Fluminense

Master Research Fellow

June 2023 - March 2024 (10 months)

Rio de Janeiro, Brasil

Laboratory for Advanced Studies in Responsive Operations and Optimization

4 years 7 months

Data Scientist

December 2020 - March 2023 (2 years 4 months)

Rio das Ostras, RJ, Brasil

- Build intelligence reports on Blowout Preventer equipment failures. Tools: Power BI, Python, ETL, IoT, Big Data, Clustering, Machine Learning, Predictive Models, Excel, SQL, MongoDB, Canva, User Experience.
- Conduct ETL processes and carry out data mining and text analysis to support project decisions. Tools: Python, Jupyter Notebook, SQL, R Studio.
- Construct the business intelligence space BOP Data Universe that brings together databases from partner companies, including Petrobras and Transocean. Tools: Power BI, Python, SQL, Jupyter Notebook, Excel.
- Maintain and update the intelligence dashboards used in the project activities. Tools: Power BI, Python, Jupyter Notebook.
- Build a high fidelity virtual physical model for equipment used in the drilling phase of underwater wells known as Blowout Preventer (BOP). Tools: Matlab, Simulink, and toolboxes like Curve Fitting, Optimization, Simscape.

Junior Data Scientist

September 2018 - December 2020 (2 years 4 months)

Rio das Ostras, Rio de Janeiro, Brasil

- Build intelligence reports on Blowout Preventer equipment failures. Tools: Power BI, Python, ETL, IoT, Big Data, Clustering, Machine Learning, Predictive Models, Excel, SQL, MongoDB, Canva, User Experience.
- Conduct ETL processes and perform data mining and text analysis to support project decisions. Tools: Python, Jupyter Notebook, SQL, R Studio.
- Create the business intelligence space, BOP Data Universe, that consolidates databases from partner companies, including Petrobras and Transocean. Tools: Power BI, Python, SQL, Jupyter Notebook, Excel.
- Maintain and update the intelligence dashboards used in project activities. Tools: Power BI, Python, Jupyter Notebook.

- Develop text mining on Blowout Preventer equipment failure records, aiming to improve understanding of maintenance activities and key failure modes observed in the industry. Tools: Python, Jupyter Notebook, NLTK, Scikit-Learn, R Studio, Power BI, and MS-Excel.

Innovative Enterprises Lab

Strategic Planning Analyst

January 2019 - June 2020 (1 year 6 months)

Rio das Ostras, Rio de Janeiro

- Develop a new strategic plan for the National Land Transport Agency (ANTT) for 2020-2030.

- Process data from interviews with the directors of the ANTT and representatives of the main stakeholders. Tools: Excel, Python, Canva, LucidChart.

- Conduct interviews with various superintendents and managers of the agency to understand the functioning and flow of the decision-making process and the responsibilities of each management and superintendent of the organization.

- Update strategic drivers, redefining the strategic map.

- Define strategic indicators, and setting goals. Build intelligence dashboards to monitor indicators associated with the agency's strategic objectives and goals. Tools: Power BI, Python, SQL, Excel, Canva.

FAPERJ

Research Fellow

May 2017 - May 2018 (1 year 1 month)

Rio das Ostras, Rio de Janeiro

- The main objective of the research was a comprehensive study of Fourier Analysis as a tool for solving PDEs (Partial Differential Equations).

- The Fourier Series and Fourier Transform were studied and this theory was applied to solve the main models of known Partial Differential Equations: heat, wave, and Laplace equations.

- It was possible to solve the well-known Black-Scholes equation for pricing options in the European market.

Education

Universidade de São Paulo

Master of Business Administration - MBA, Data Science & Analytics · (June 2025 - December 2026)

Federal Fluminense University

Master degree in Production Engineering and Computing Systems, Implementation, Management and Evaluation of Production Systems · (April 2021 - April 2024)

Udacity

AWS Machine Learning Engineer, Machine Learning · (October 2022 - February 2023)

Udacity

Data Scientist, Data Science · (September 2021 - January 2022)

Federal Fluminense University

Bachelor degree, Production Engineering · (2015 - 2020)